

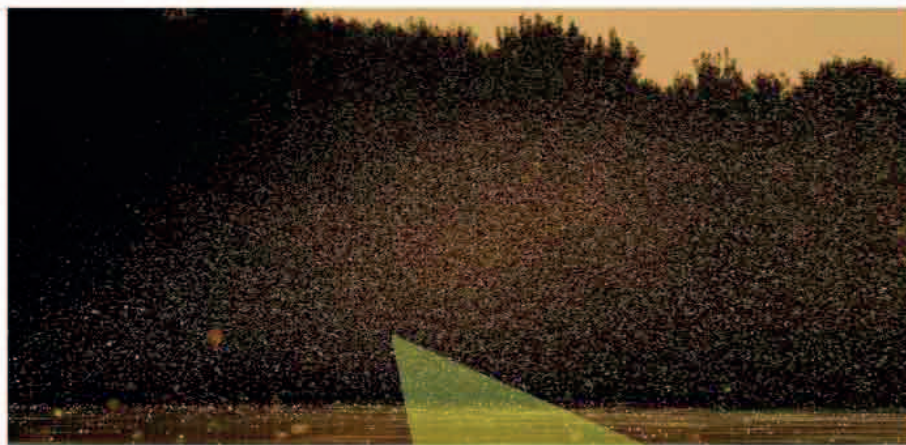


Freshwater life

Most ecosystems are complex, but freshwater habitats, such as rivers, lakes, and marshes, are often easy to understand. In a lake, for example, plants and algae are the producers, while zooplankton, insects, fish, and large, fish-eating animals are the consumers. But the particular species in each ecosystem depend on whether the water is flowing or still, warm or cold, and so on.

RICH AND POOR

The water of upland lakes and streams is often almost pure, with few dissolved nutrients. This limits the growth of plankton, but the cool water is often rich in oxygen, which suits insects and fish. Lowland waters, such as this lake, have more nutrients, which encourages plankton. But the warmer water contains less oxygen, which can be a problem for animals.



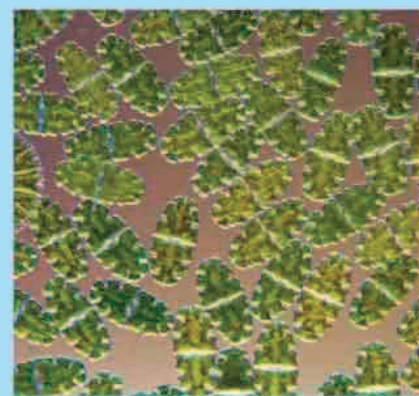
DRIFTING SWARMS

The algae drifting in fresh water are eaten by the tiny animals of the zooplankton—creatures such as water fleas and copepods—as well as various aquatic insects. These include the young of winged insects, such as mayflies, which emerge as flying adults in spring. In some regions, mayflies gather over the water in spectacular swarms.



VITAL ALGAE

On land, the primary producers are plants, which capture sunlight and use it to turn air and water into food. In rivers and lakes, this is the job of microscopic single-celled algae that drift in the water as plankton. These live in all freshwaters that contain enough nutrients to support them. They are the first link in the aquatic food chain.



▲ MICROORGANISMS

Although they are not plants, these microscopic freshwater algae make food in the same way.